Ch. 5.1: The Single Market.

















Features of the Single Market.

- **Single Market:** Stimulation of industrial and commercial expansion within a large, unified economic area.
- Single European Act July 1987 - the biggest achievement of the EU.
- The Single Market is a frontier-free area within which
  - ***people***,
  - ***goods,***
  - ***services***, and
  - ***money*** can all move around freely.
- **Specific achievements:**
  - elimination of all border controls on goods and customs controls on people,
  - barrier free travelling of people (Schengen Agreement),
  - mutual recognition of technical standards and national rules,
  - liberalisation of the services sector (banking, insurance, telecommunications, air traffic etc.) leading to lower prices,
  - mandatory competitive bidding for large projects,
  - mutual recognition of vocational and educational qualifications ("Bologna Process" involving now 49 countries).

Impact of the Single Market.

- **Massive increase of standards of living in EU.**
- More choice for consumers and business.
- Change in industrial production structures: new factories in the new Member States, Porsche Cayenne produced in Bratislava.....
- EU is the third largest economy of the world.
- Increase of exports from highly competitive states like Germany to other Member States created however massive economic and financial imbalances within the EU.
- Liberalization of banking (following global liberalization) contributed to financial crisis of 2009.
- Brain drain in less developed regions of the EU (e.g. doctors).
- Economic migration: low income workers move from CEE to DE, AT, NL, DK, SE…….
  - A major argument in UK in the BREXIT discussions (3 million workers from CEE).
  - A discussion item in the new trade arrangement between Switzerland and the EU.
- **Note:** EU not yet a single economic area, because some sectors of the economy (public services) are still subject to national laws and the individual EU countries still largely have the responsibility for taxation and social welfare.

Products of the Single Market.

- The Single Market has enabled a highly successful cooperation between Member States:
  - Airbus has become together with Boeing (USA) the world's leading aircraft manufacturer: in total 18.000 aircrafts produced.
  - Volkswagen Group became the second largest car maker in the world with 9 million cars sold in 2025 after incorporating SEAT, Skoda, Porsche, Bentley, Lamborghini, Bugatti.



World's largest passenger airplane A380 built by European corporation Airbus.

Impact of the Single Market.

Source: Facebook 2024



Italy 1970



Italy 2023

Policies Underpinning the Single Market:
Transport.

- **Free access to the international transport market:**
  - Flixbus, Uber…..
  - Skies over Europe have been liberalized with the result that there is greater scope for capacity-sharing between major airlines, reciprocal market access and the freedom to set fares.
- **Trans European Networks (TEN):**
  - Part of the new program on „Connecting Europe Facilities“.
  - Transport – TEN (30 billion)
  - Energy (9 billion) and
  - Digital networks (3 billion)



The Connecting Europe Facility (CEF):
Highways.

EU Highway System:
60.000 km
Backbone of
European transport.



Source: Wikipedia 2024

- **Major challenges to transport:**
  - Global Climate Change Agreement (Paris 2015) requires a reduction of CO_2 emissions by 90% by 2050.
  - Share of road traffic in GHG emissions now ca 20%: in theory traffic will have to be CO_2 emission free by 2050.
  - Reduction of mobility per se, incl. use of modern communication tools.
  - Replacement of combustion engines by electrical motors.
  - Change of transport modality mix from individual cars to mass traffic and bicycles.
  - Mid-distance air travel replaced by fast trains.

The Connecting Europe Facility (CEF):
High Speed Train Network.



10.000 km in 2024
at speeds of > 200km/h

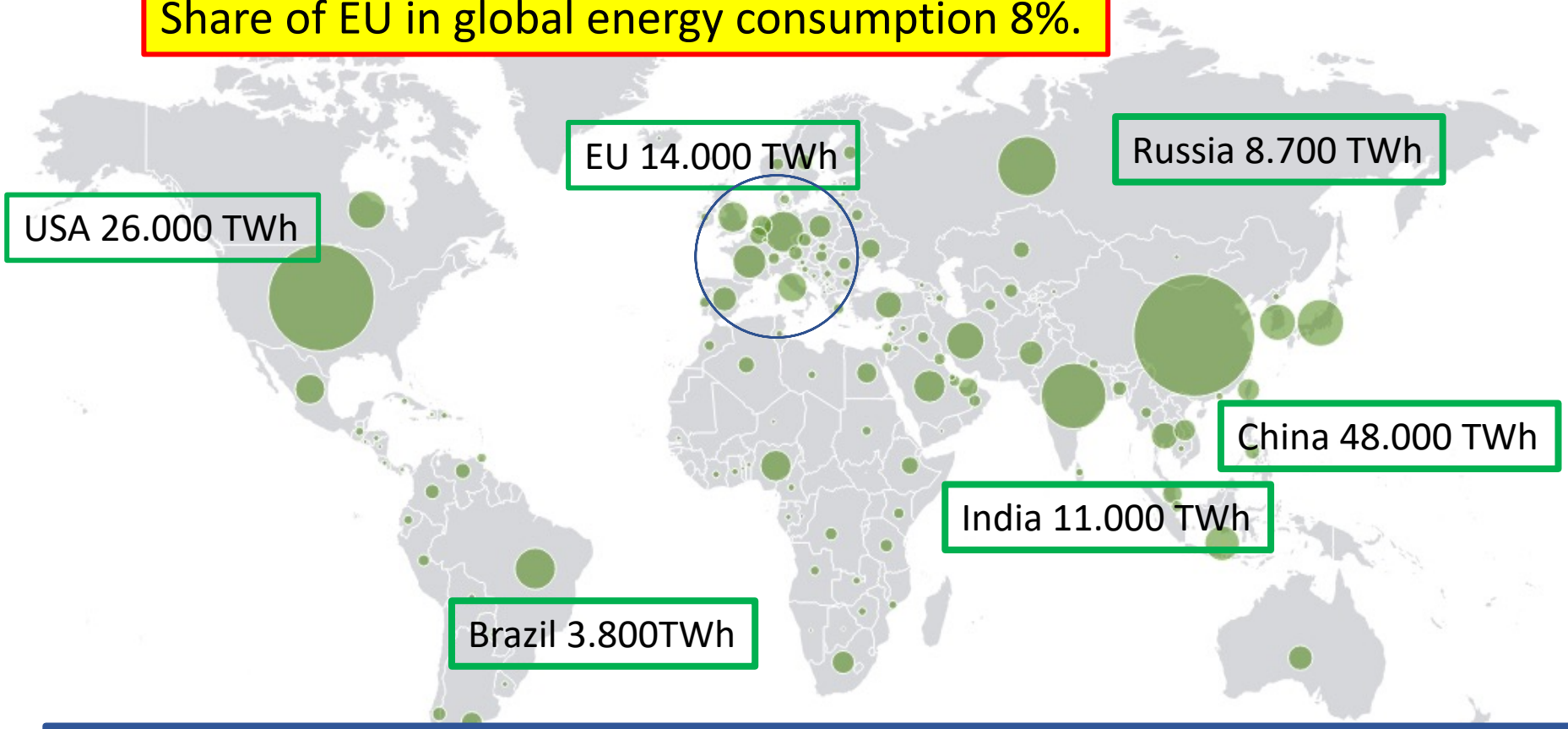
- Inefficient patchwork: 11.000 national rules in existence.
- Reliability issues:
  DB 2 million changes in schedules in 2023.

Source: European Commission, EU Court of Auditors

Policies Underpinning the Single Market:
Energy.

Total Global Annual Energy Consumption is 164.000 TWh (2024).

Share of EU in global energy consumption 8%.

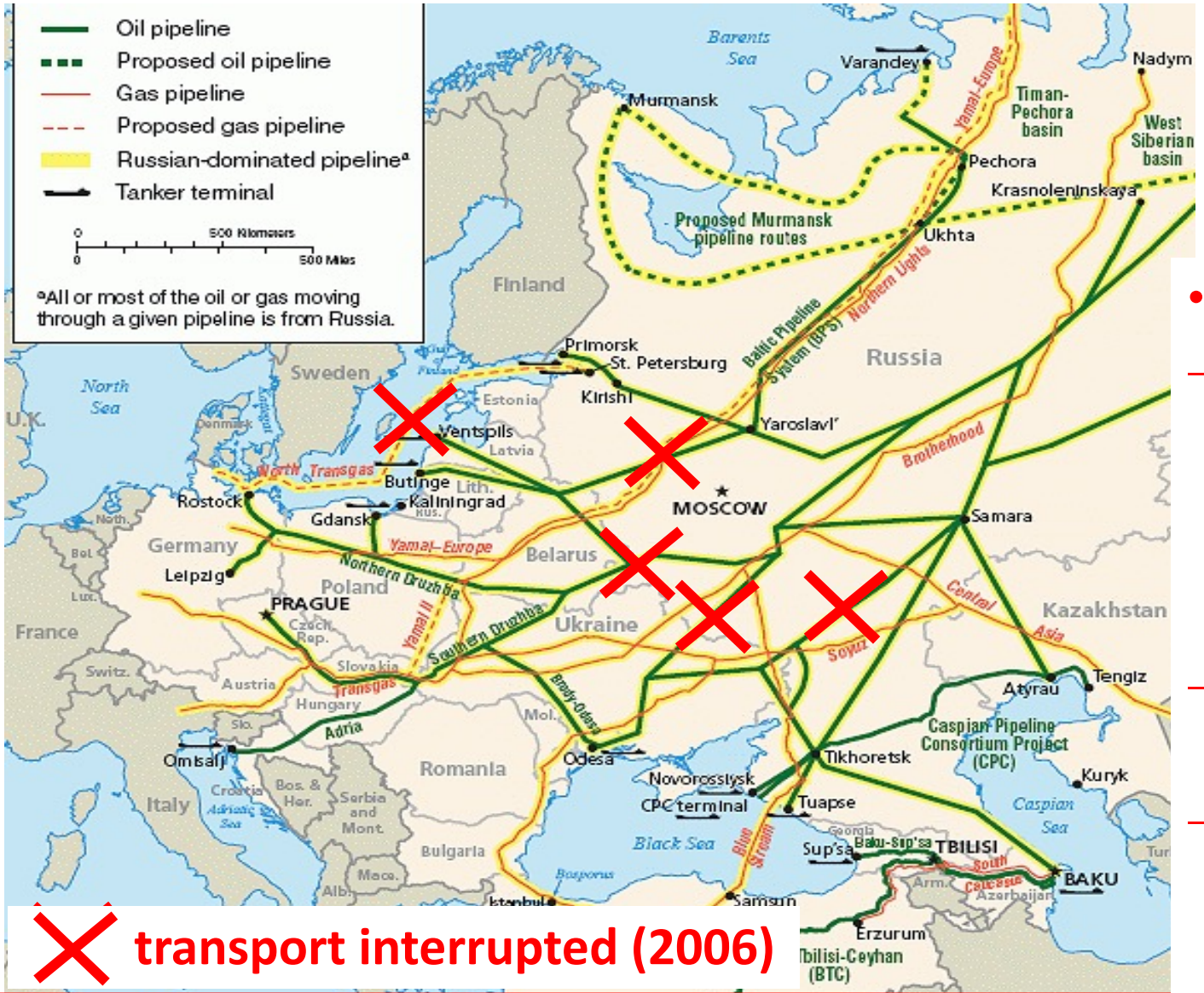


- **The EU is in global terms a major consumer of energy.**
- **Energy sources:**
  - oil 37,3%,
  - coal 9,6%,
  - natural gas 20,4%,
  - nuclear 11,8%,
  - renewables (hydro, bio, wind, solar) 19,5% (2023).

The "Green Deal" of 2019 is even aiming at "zero-carbon emission" – meaning a 90% reduction of Green House Gas Emissions by 2050 (now 67% of energy in the EU is produced from fossil fuels).
Major challenge is defossilisation of energy supply.

ce: https://ourworldindata.org/energy-production-consumption

EU Energy Security.



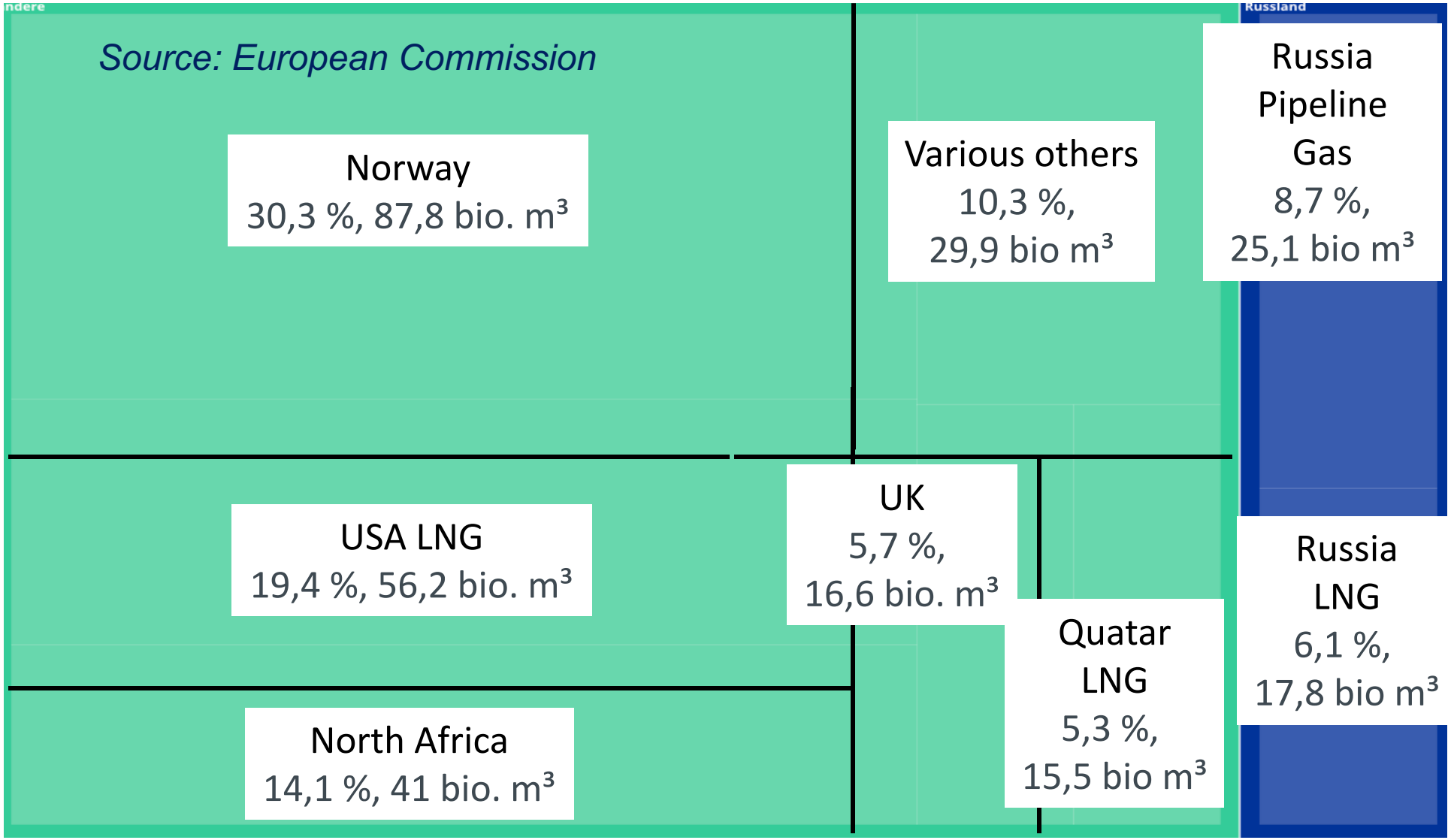
Oil and gas pipelines for energy supply of the EU (2000).

- The EU imports 57 % of its energy (82% of its oil, 57% of its gas and 97,5% of its uranium demands) at a cost of 1 billion EUR per day.

- **Problems:**
  - The security of energy supply is now a severe issue:
    Middle East unstable,
    Russia – massive reductions due to conflict and sanctions.
  - Important gas pipelines destroyed or interrupted.
  - Long term goal: Substitution of oil and gas by RE – problem of volatility of wind and solar energy.
  - Massive increase in costs of energy supply.

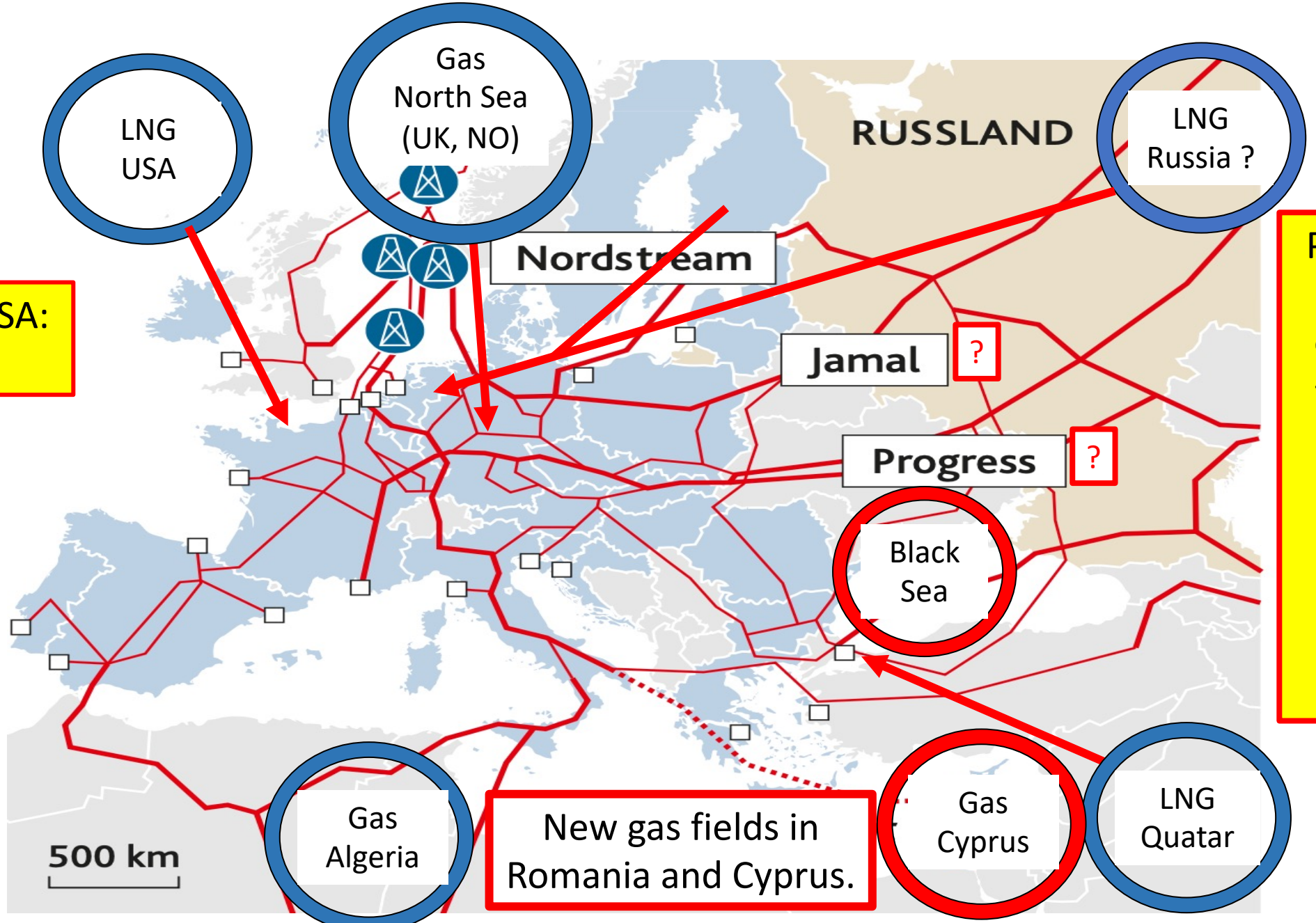
EU Energy Situation:

Supply with Natural Gas 2023. Total 289,9 billion m³



Share of Russia has dropped from 40 to 15% (2021-2023).

Future Supply of EU with Natural Gas.



LNG from USA: price?

Potential of Russia to deliver gas to the EU? Sanctions from EU. Also transition stop of Ukraine.

New gas fields in Romania and Cyprus.

Transformation of Electricity Production.

Electricity Production in EU 2.600 TWh/a (2024):

- **CO_2-emission free production 71%**
  - 25% nuclear
  - 14% hydro
  - 3% biomass
  - 19% wind
  - 10% solar
- **fossil production 29%**
  - 14% natural gas
  - 10% coal
  - 3% oil
  - 2% garbage

stable supply

volatile supply

stable supply

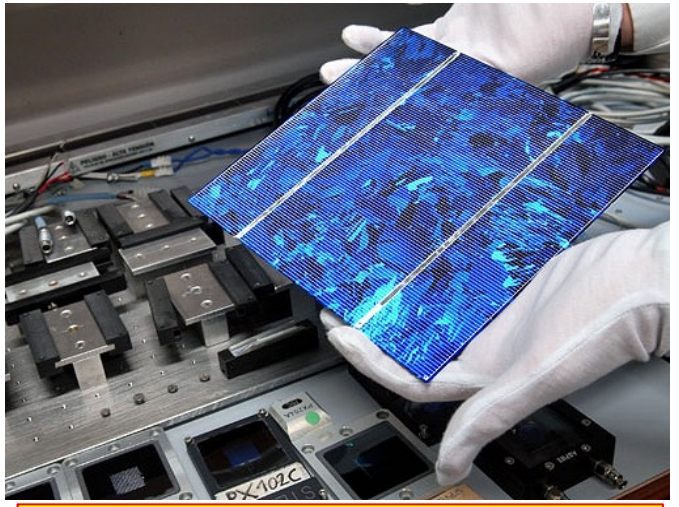


Enercon E-147: 5 MW,
height 155 m,
rotor 62 m
Av. Yield in EU: 26%
Production 10 GWh/a.

- **Substitution of fossil fuels by wind (50%) and solar (50%) energy:**
  - 50.000 new wind turbines (now 130.000).
  - 5.000 km² new photovoltaic panels (now 2.000 km²) .
  - Tens of thousands kilometer new power lines.
  - Costs: 3.000 billion EUR (?)

58% of EU electricity production then volatile.

PV-modules:
polycristalline silicon.

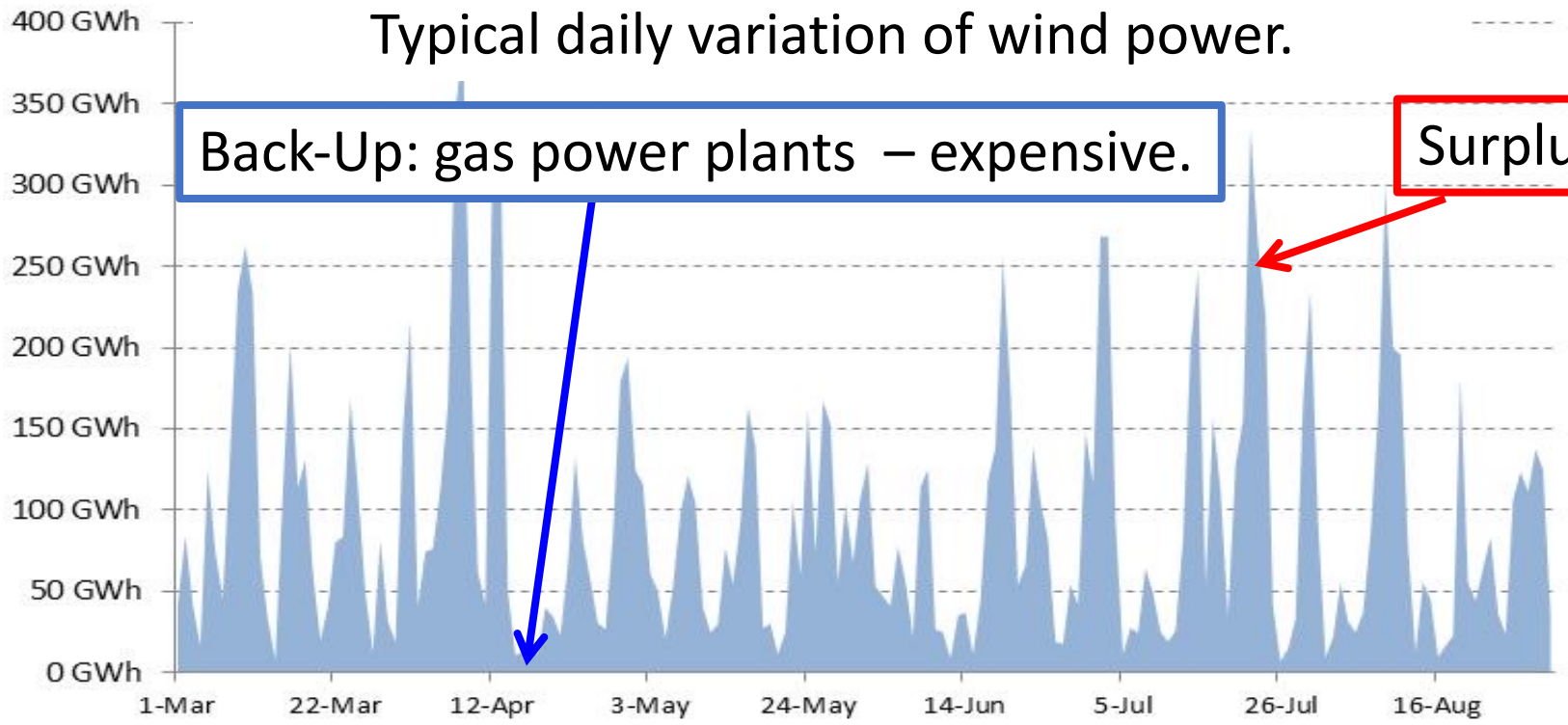


Av. yield in EU is 12% of nominal peak output.
Production of 1 TWh/a needs 10 km² PV panels.
90% imports from China.

Source: Statistica 2025

Transformation of Electricity Production.

- ***Problem 1: Electricity from wind and sun is volatile.***



Transformation of electricity production in Austria:
100 billion EUR till 2030.

- ***Problem 2: Use of surplus power.***

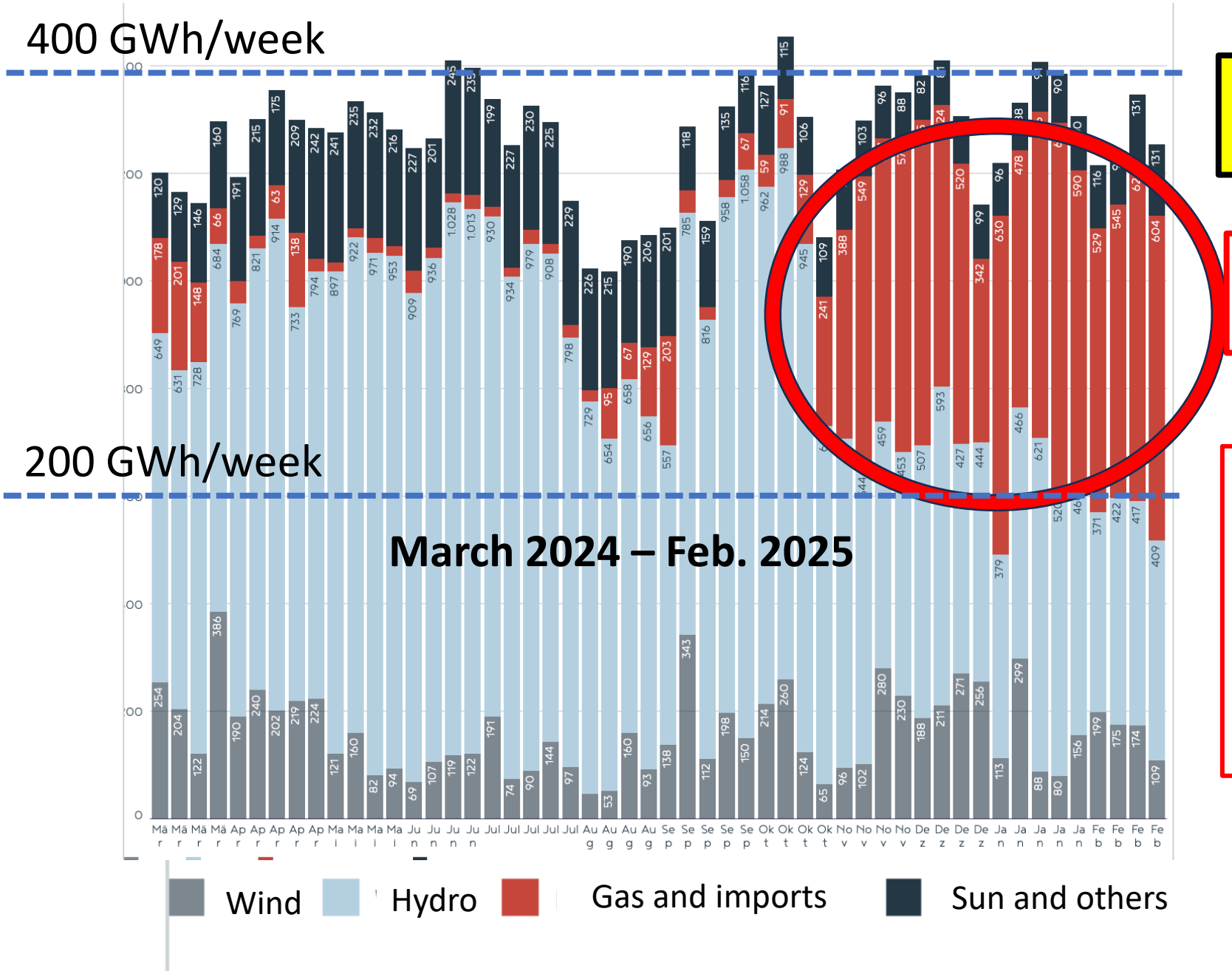
– Storage in hydro pump power plants (Kaprun) – insufficient capacities.

– Production of „Green Hydrogen“ with electrolysis of water or Syn-Fuel with CO_2 und H_2:

➢ Low energy density of H_2 (30% of natural gas) and at least 5x more expensive.

➢ CO_2 from cement, in development, high price of Syn-Fuel (5-10x higher).

Seasonal Volatility of Renewable Energies.



Electricity consumption 2024 in Austria (GWh/week).

Winter power gap:
gas power plants, imports

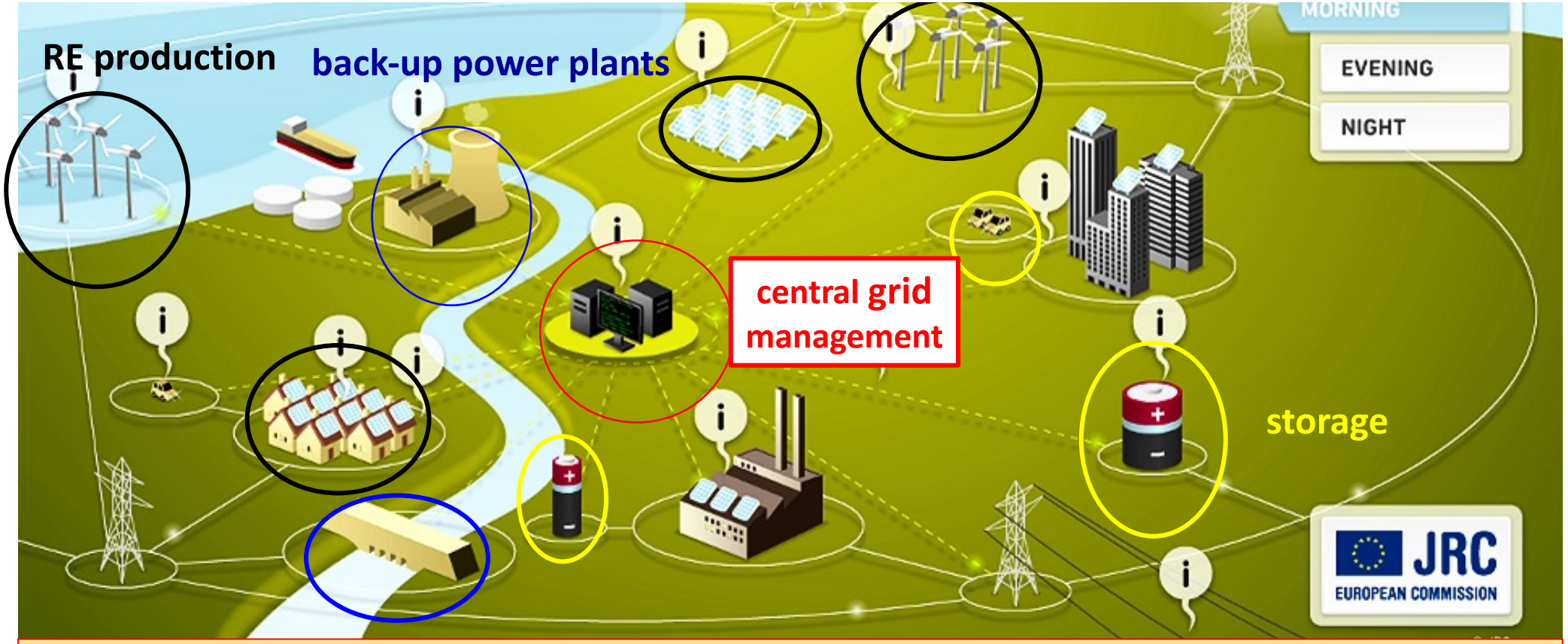
Electricity production 2024:
81,5 TWh
of this
57 TWh Renewable Energy –
hydro, wind, sun, bio
(70% of total production)

https://www.apg.at/infografiken

„Smart Grids“.

“Intelligent” network of electricity production, storage and consumption through local, regional and continental grids.

Smart meters enable production oriented supply and consumption oriented pricing.



Hugely complex and expensive system:

Enormous technical and sociological challenge for the European Union.

Investments in Austria estimated 20 billion EUR (EU 1.000 billion EUR?).

Source: JRC 2012

Policies Underpinning the Single Market:
Competition.

- **Robust competition policy vital for free trade within the European single market.**
- Concerns agreements between businesses, aid from public authorities or any unfair monopoly.
  - Any agreement falling under the Treaty rules must be approved by the Commission.
  - Any merger or takeover that could lead to a company having a dominant position in a particular sector must be notified to the Commission.

- **Competition in air transport:**
  - The EU has enabled airlines to compete against each other in 1980 and to merge companies creating a huge single market for air travel.
  - The result has been the establishment of 3 major airline operators, led by British Airways, Lufthansa, Air France-KLM and the launch of dozens of new airlines, including 'no-frills' or 'low-cost' airlines (Ryan, Easyjet, Wizz…).
  - Any airline granted a license in an EU country can operate air services anywhere in the EU, including domestic services.
  - More competition led to new routes, more flights and cheaper air travel for all.

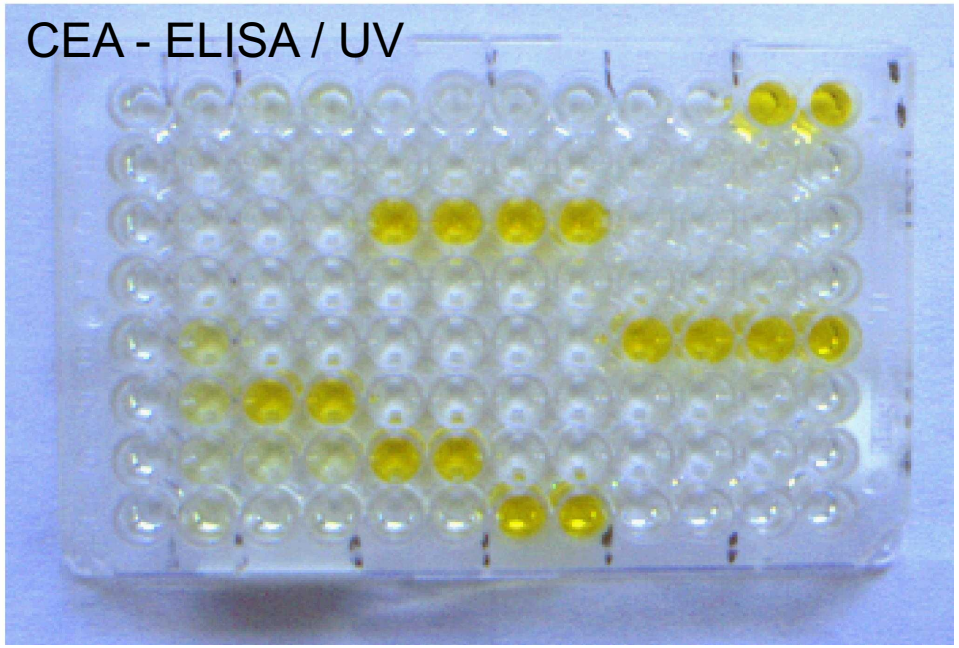
Policies Underpinning the Single Market:
Consumer Protection.

- **Europe is a huge single market for consumer goods.**
- Consumers want to be sure that the products they buy are high quality and safe.
- European product safety rules and high quality product standards apply to all member states.
- A rapid information exchange service (RAPEX) has been established to notify the public about a product problem and if necessary a product recall can be started.
- An example is the ban on the use of phthalates:
  - Phthalates are chemicals which were used in toys and other childcare products to make plastics softer.
  - But they were also associated with potential health risks for children.
- These types of decisions are based on careful scientific assessment.
- **Achievements:** European Health Insurance Card, European Chemicals Legislation (REACH), prohibition of unjustified health claims in food, TSE(BSE) action

The TSE (BSE) Crisis of 1999.



- TSE = Transmissible Spongiform Encephalopathies:
  Bovine Spongiform Encephalopathy („mad cow disease“)
- Occurred in UK: First warning signs ignored.
- Finally nearly 200.000 cases of BSE discovered.



- Over 100 cases of variant CJD in the EU with estimates ranging up to thousands of potential human victims.
- Five million cattle destroyed in UK.
- Economic loss in UK 10 billion EUR.
- Breakdown of beef market:
  billions of EUR for support measures.

- Protective measures introduced throughout the EU:
  - Elimination of risk materials.
  - Testing of cattle for BSE slaughtered above 30 months, applied to up to 10 million slaughtered animals every year.

How Much Protection is Too Much?

- **EU regulation in consumer protection matters is often seen as negative by the general public:**
- ***„The European Union is invading the privacy of the citizen.“***

- Limitation of the output of music players to reduce the danger of damaging hearing of young people ?
- Strong warning on the health danger associated with the consumption of alcohol by labelling bottles in a similar way as cigarette boxes („Consumption of alcohol seriously damages your health“ on a bottle of Chateau Lafitte) ?



- Excessive warning on cigarette boxes?
- „Traffic light system“ labelling of healthiness of food stuff?
- „Dangerous sofas Directive“?
- Campaign „Europe acts against obesity“?
- Measures to reduce resource consumption – cheap clothing, food, esp. meat?

The Economic Status of the European Union.

- **Gross domestic product (GDP):**
  - One of the ways of measuring the size of the economy of a country, invented by Simon Kuznets 1934 (Nobel Prize 1971).
  - GDP is defined as the total market value of all final goods and services produced within a given country in a given period of time (usually a calendar year).
  - It is also considered the sum of value added at every stage of production (the intermediate stages) of all final goods and services produced within a country in a given period of time, and it is given a money value.
- **The most common approach to measuring and understanding GDP is the expenditure method:**
  - *GDP = consumption + gross investment + government spending + (exports − imports)*, or, GDP = C + I + G + (X-M)
- Cross-border comparisons can be made on the basis of the purchasing power parity (PPP) method or current exchange rate method.

The Economic Status of the European Union (2023).

How wealthy is the EU compared to the rest of the world?

GDP per capita:

Luxemburg: 109.00 EUR

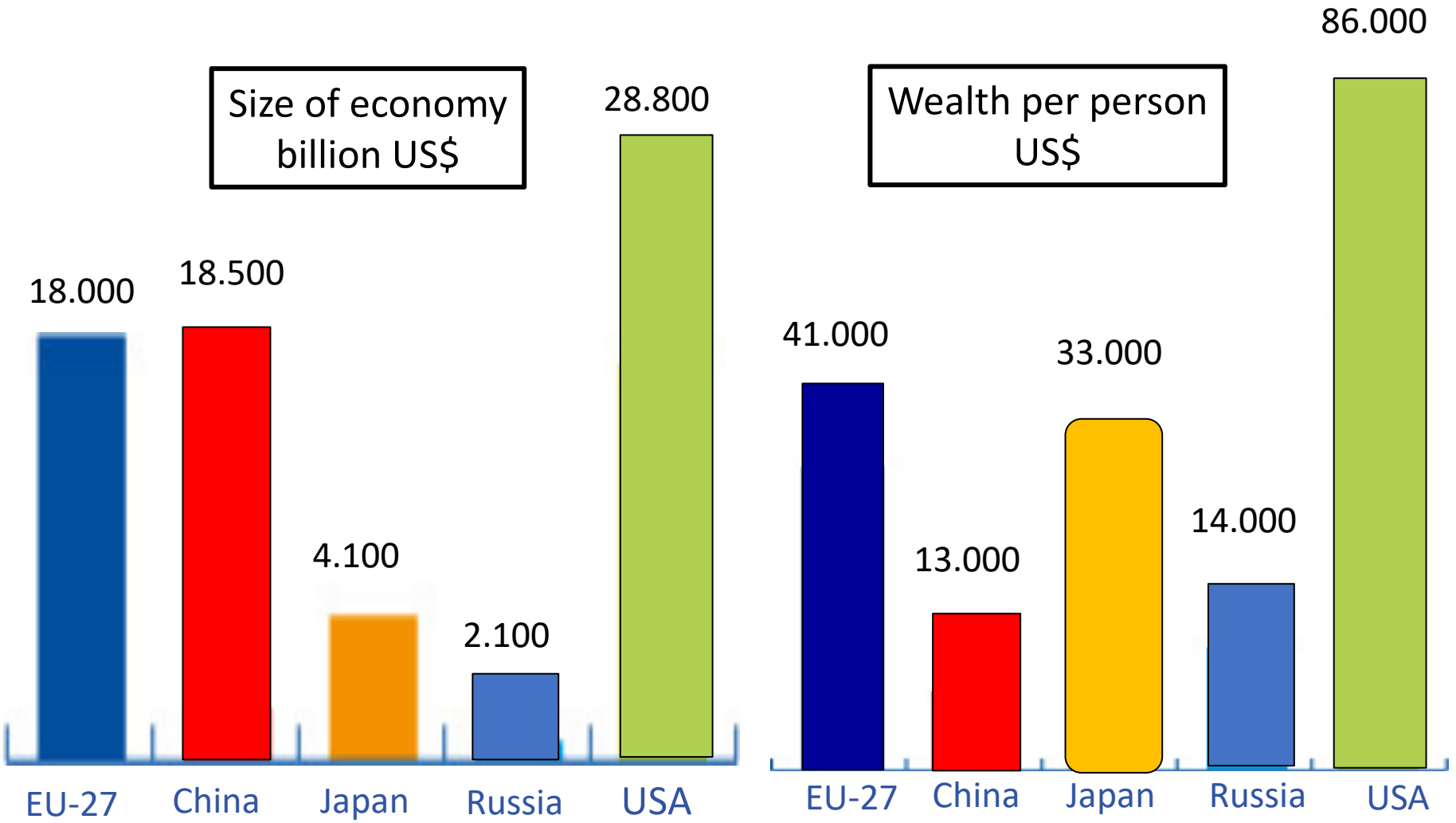
Germany: 43.200 EUR

Austria: 44.800 EUR

Poland: 28.300 EUR

Romania: 26.000

Bulgaria: 19.800 EUR



GDP growth rate:
0,5 - 2%

Labour force by sector (2021):
70% services
25% industry
3% agriculture

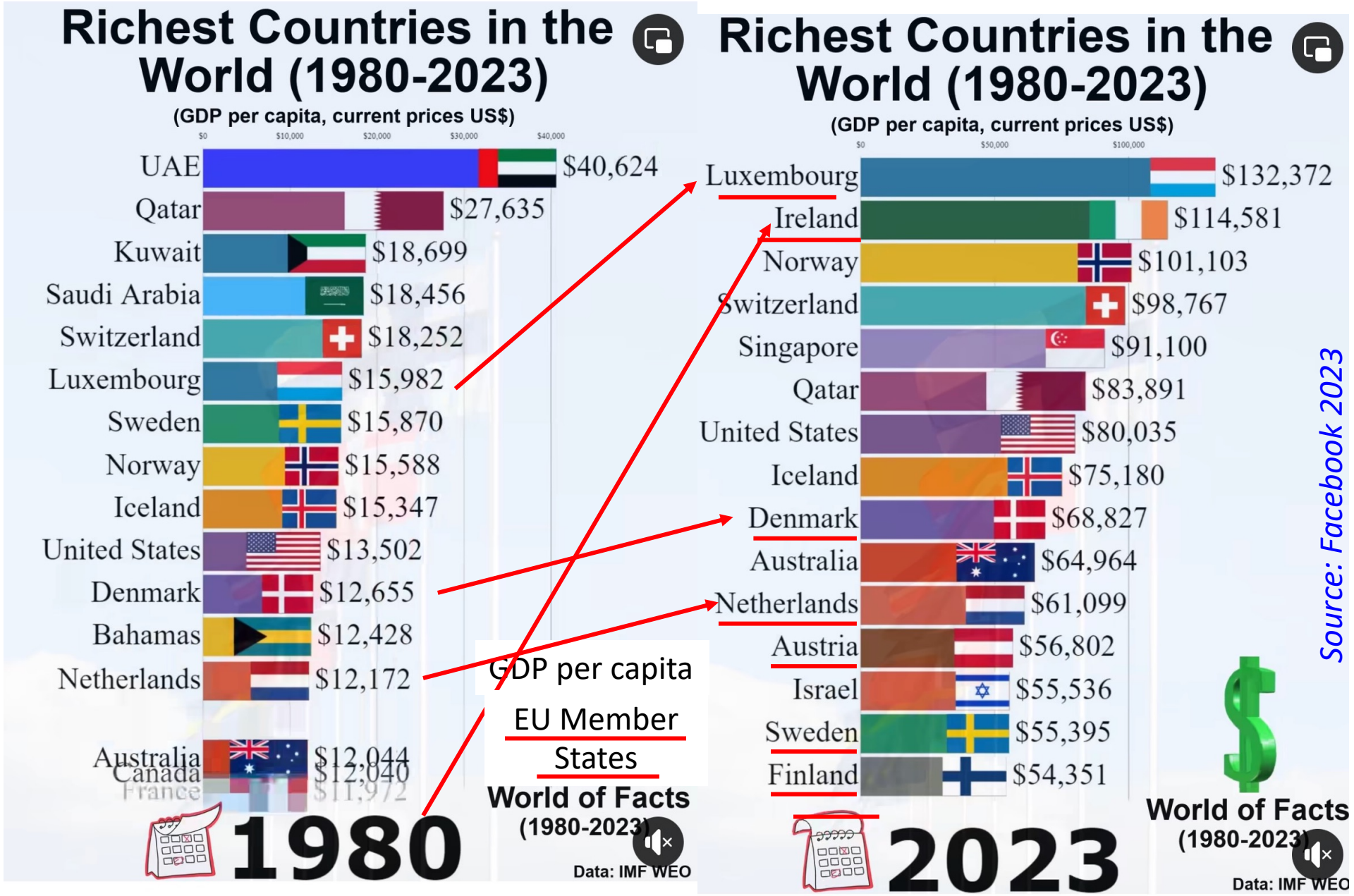
GDP by sector:
70% services
25% industry
3% agriculture

Nominal Gross Domestic Product based on current prices for 2023.

Nominal Gross Domestic Product per capita.

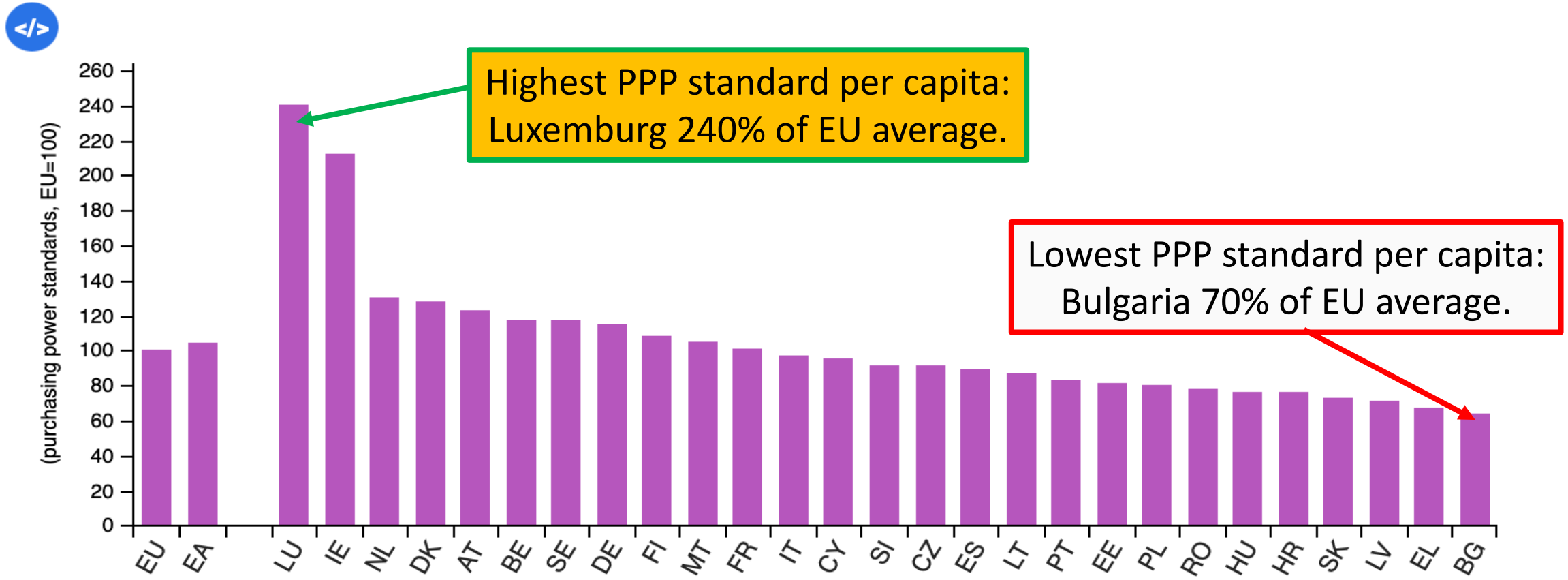
Source:Forbes 2024

The Evolution of the Economic Status of EU Member States.



The Economic Status of Member States in the European Union.

(purchasing power standards, EU indexed at 100)



Highest PPP standard per capita: Luxemburg 240% of EU average.

Lowest PPP standard per capita: Bulgaria 70% of EU average.

Ratio richest/poorest Member State in per capita PPP considering actual buying power: 3,5:1

Major policies and activity areas of the EU:
Economic and Social Solidarity.

- The Single Market has been a huge success in the EU, but very large differences exist in the economic situation of different Member States reflecting their
  - history,
  - geographical position,
  - development priorities,
  - attitude of the people,
  - quality of the governance system…….
- Therefore Europe-wide free competition has been complemented with Europe-wide solidarity:
  - Regional Policy
  - Common Agricultural Policy
  - Common Fisheries Policy
  - Environmental Policy
  - Social Policy